

University of Central Punjab

Faculty of Information Technology and Computer Science

Course Title: Software Construction & Development

Assignment 03

Scenario: Parking Fee Calculator using Java Swing

Name:		Registration No.:	
Section:		Date:	
Course Code:	SESD3242	Total Marks:	50
Deadline:	06-06-2026	Submission:	Typed / Printed

CLO#	Course Learning Outcome (CLO)	Taxonomy Level	PLO
CLO2	Apply software design principles, refactoring techniques and design patterns to develop collaborative, maintainable, and high-quality Java applications.	C3	2

Instructions:

1. Late submission will cause 20% deduction. No submission after 2 days.
2. Assignment should be properly typed and formatted.
3. Attach this title page to your submission.
4. Evaluation may be viva-based. Students should understand their code.
5. Plagiarism is not allowed. Write and explain your own solution.

Question 1:

Write backend code for the following GUI using Java Swing.

Scenario

A user enters the number of parking hours used. When the Generate Fee button is clicked:

- If hours are less than or equal to 3 -> Rs. 50 per hour
- Otherwise -> Rs. 80 per hour
- Display the total parking fee in the output field.

Handle:

- NumberFormatException using try-catch
- Empty input field validation
- Use finally block to show operation completion message

Suggested GUI Layout:

Parking Fee Calculator	
Enter Parking Hours:	e.g. 4
Total Fee (Rs.):	---
[Generate Fee]	[Reset]
Required Components:	JTextField, JButton, JLabel, JOptionPane

Expected Functional Requirements:

Feature	Expected Behaviour	Marks
Input	Accept parking hours from user.	5
Calculation	Apply correct rate based on hours.	10
Exception Handling	Handle invalid number and empty field.	10
finally block	Show completion message after try-catch.	5
Reset Button	Clear input and output fields.	5
Clean Code	Use readable variable names and proper formatting.	10
Title Page / Formatting	Attach title page and submit typed work.	5